

# Which Error Can the District Afford?

Topic 6.7 · TODAY'S PROBLEM

A district tests  $H_0 : p = 0.05$  versus  $H_a : p > 0.05$ , where  $p$  is the late-alert proportion. Both plans use  $n = 200$  and have the table's operating probabilities. Safety prioritizes avoiding retention when  $p = 0.10$ ; Budget prioritizes avoiding replacement when  $p = 0.05$ . No common cost scale exists.

**Dev:** "Use A. Smaller  $\alpha$  lowers both error risks."

**Lin:** "Use B. Its higher power has no cost when safety matters."

Test plans ( $n = 200$ )

| Plan | $\alpha$ | Power at $p = 0.10$ |
|------|----------|---------------------|
| A    | 0.010    | 0.748               |
| B    | 0.050    | 0.877               |

## FIRST TAKE (5 min, on your sheet)

Which plan would you try first? Name the office you prioritized and one competing cost.

- DOK 1** (a) Describe a Type I error and a Type II error in this context. Identify what  $P(\text{reject } H_0 \mid p = 0.10)$  represents.
- DOK 2** (b) For each plan, calculate  $\beta$ . Then give the expected number of Type I errors per 1,000 repeated tests when  $p = 0.05$  and Type II errors per 1,000 repeated tests when  $p = 0.10$ .
- DOK 3** (c) Adjudicate both claims. Explain why no plan is universally best; recommend one for a stated office using both error probabilities and their per-1,000 consequences. Name information needed for a district-wide choice and a change that improves power at fixed  $\alpha$ .

Video + follow-along: [u6\\_lesson7\\_live.html](https://u6_lesson7_live.html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

